

# SALUS

SALUS - The evidence behind every path to healing.

SALUS is a monorepo for an evidence-first medical comparison platform spanning Allopathy, Ayurveda, Siddha, and Naturopathy. The current codebase provides a working frontend, API, shared domain validation package, and AWS SAM infrastructure baseline.

## Current implementation status

Implemented now:

- React + Vite web app with:
  - Dashboard route and add-evidence route
  - evidence table view (query + render)
  - evidence submission form
- Hono API with:
  - `GET /health`
  - `GET /conditions`
  - `GET /evidence?conditionId=...`
  - `POST /evidence`
- Shared domain schemas via Zod for:
  - medical systems
  - A/B/C evidence grades
  - registry identifier prefix validation
- Local data behavior:
  - in-memory fallback for local dev
  - DynamoDB persistence when `DYNAMODB_TABLE` is configured
- AWS SAM baseline for:
  - Lambda function URL API
  - DynamoDB table + condition GSI
  - Cognito resources
  - S3 + CloudFront frontend hosting

Planned/in-progress (not fully implemented in app runtime yet):

- Editorial publish workflow and API auth enforcement
- Automated ingestion pipelines (PubMed, ClinicalTrials.gov, AYUSH/DHARA, CTRI/ICTRP)
- Pramana score computation and calculus intelligence modules
- Full free-tier guardrail resources in SAM (budget alarms, WAF policies, extended cost controls)

See roadmap details in [docs/IMPLEMENTATION\\_PLAN.md](#).

## Stack

- Frontend: React 18, TanStack Query/Router/Table, Vite, Vitest
- API: Hono, Node.js 20, AWS Lambda-compatible handler
- Domain package: Zod schemas and shared types
- Data: DynamoDB (plus local in-memory fallback)
- Infra: AWS SAM (Lambda, DynamoDB, Cognito, S3, CloudFront)

## Repository layout

- [apps/web](#): frontend app
- [services/api](#): API service
- [packages/domain](#): shared schemas/types/validation
- [infra](#): SAM template
- [docs](#): architecture, guardrails, deployment, implementation plan, math spec

## Prerequisites

- Node.js 20+
- npm 9+

For cloud deployment:

- AWS CLI configured
- AWS SAM CLI

## Quick start (local)

1. Install dependencies.

```
npm install
```

2. Start web + API in parallel.

```
npm run dev
```

3. Open:
  - Web: <http://localhost:5173>
  - API: <http://localhost:8787>

Notes:

- API runs without AWS credentials in local mode using in-memory storage unless `DYNAMODB_TABLE` is set.
- Frontend API target defaults to `http://localhost:8787` and can be overridden with `VITE_API_BASE_URL` .

## Scripts

Root scripts in [package.json](#):

- `npm run dev` : run web and API concurrently
- `npm run build` : build domain, API, web
- `npm run test` : run all workspace tests
- `npm run lint` : type-check lint gates across workspaces
- `npm run docs:pdf` : generate both calculus spec PDFs
- `npm run docs:pdf:academic` : generate academic PDF variant
- `npm run docs:pdf:executive` : generate executive PDF variant

## API summary

Current routes in [services/api/src/app.ts](#):

- `GET /health`
- `GET /conditions`
- `GET /evidence`
- `POST /evidence`

Input validation uses `treatmentEvidenceSchema` from [packages/domain/src/index.ts](#).

## Evidence model guardrails (current)

Current schema guardrails include:

- Evidence grade constrained to `A | B | C`
- Registry identifier prefix check for:
  - PMID
  - DOI
  - CTRI
  - ICTRP
  - AYUSH
  - DHARA
  - NCT
- URL shape validation for source links
- Contraindications and interaction warnings typed as arrays

## Documentation map

- Architecture vision: [docs/ARCHITECTURE.md](#)
- Implementation roadmap and phased delivery: [docs/IMPLEMENTATION\\_PLAN.md](#)

- AWS deployment steps: [docs/DEPLOY\\_AWS.md](#)
- Cost guardrail policy: [docs/ZERO\\_COST\\_GUARDRAILS.md](#)
- Calculus specification (source): [docs/CALCULUS\\_MATH\\_SPEC.md](#)
- Calculus specification (PDF): [docs/CALCULUS\\_MATH\\_SPEC.pdf](#)
- Calculus executive PDF: [docs/CALCULUS\\_MATH\\_SPEC.executive.pdf](#)

## Build calculus PDFs

```
npm run docs:pdf
```

This regenerates both PDF variants from the markdown math spec using the repository print CSS profiles.

## Deploy to AWS

Use [docs/DEPLOY\\_AWS.md](#) for the full sequence.

Minimal flow:

```
npm install
npm run build
sam build -t infra/template.yaml
sam deploy --guided
```

Then upload frontend assets to S3 and invalidate CloudFront.

## Safety and scope disclaimer

This repository is a software platform for evidence organization and presentation. It is not a clinical decision system, does not provide medical advice, and must not be used as a substitute for qualified medical judgment.